RETRACE: MODEL-RESPONSE WATERMARKING FOR DIFFUSION LANGUAGE MODELS

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ABSTRACT

Existing language-model watermarks are detected from token patterns. We detect ours from *how a diffusion language model responds when the generated text is partially masked and reconstructed*. Because such a model is trained to recover masked tokens from arbitrary surviving context, a finished text can be pushed back through the operator that produced it: corrupted with a secret key, re-denoised, and its response read. ReTrace places a watermark in that response. A keyed pair of context ablations induces a per-position log-likelihood contrast, and the watermark asks the sign of that contrast to agree with a keyed orientation, giving a test statistic that is a sum of bounded indicators with a finite-sample exact Binomial$(n, \frac{1}{2})$ null. Embedding never overrides the model: decoding follows the reference block schedule, and the watermark only decides whether a sampled proposal is emitted or redrawn. We also report the negative result that motivated this design — post-hoc token substitution is priced out on a well-trained diffusion language model, and, because both are governed by the sharpness of the same conditionals, *improving generation quality reduces post-hoc watermarkability*.

1 INTRODUCTION

Existing language-model watermarks encode provenance in *what tokens are produced*. Red–green schemes bias the token distribution toward a keyed vocabulary partition (Kirchenbauer et al., 2023); distortion-free schemes fix the sampling randomness with a keyed pseudorandom sequence; and the watermarks designed for diffusion language models exploit the extra freedom those models expose — unresolved context, per-step sampling randomness, global sequence statistics, or the order in which positions are unmasked (Hong & No, 2026). Different as they are, every one of them leaves the verifier reading a *surface signature*: a hash of token identities, a frequency, a rank. We ask a different question.

Can a watermark be detected from how the model reconstructs a text, rather than from the tokens the text contains?

A diffusion language model makes this question answerable in a way an autoregressive model does not. Its training objective is to recover masked tokens from arbitrary surviving context, so any finished text can be pushed *back* through the operator that produced it: corrupt it, re-denoise it, and observe the response. That response is a property of the text's relationship to the model, not of its symbols, and it is available to anyone holding the model — but only interpretable by someone holding the key that chose the corruption.

We call the resulting scheme **ReTrace**. A secret key derives two corruption patterns over the context of a span. Reading the model's log-probability of each *observed* token under both corruptions gives a contrast

$$g_i(y_i) \;=\; \log p_\theta\big(y_i \mid C_i^1(y)\big) - \log p_\theta\big(y_i \mid C_i^0(y)\big)\,, \tag{1}$$

and the watermark asks not that y_i belong to a keyed vocabulary subset, but that the sign of $g_i(y_i)$ agree with a keyed orientation. The evidence lives in the model's behaviour.

Getting the embedding right took one full negative result (§5). Substituting tokens — the obvious embedding — fails because a well-trained denoiser's conditionals are sharp: the median position admits exactly one token inside any usable fluency budget, and the effect *strengthens* as generation

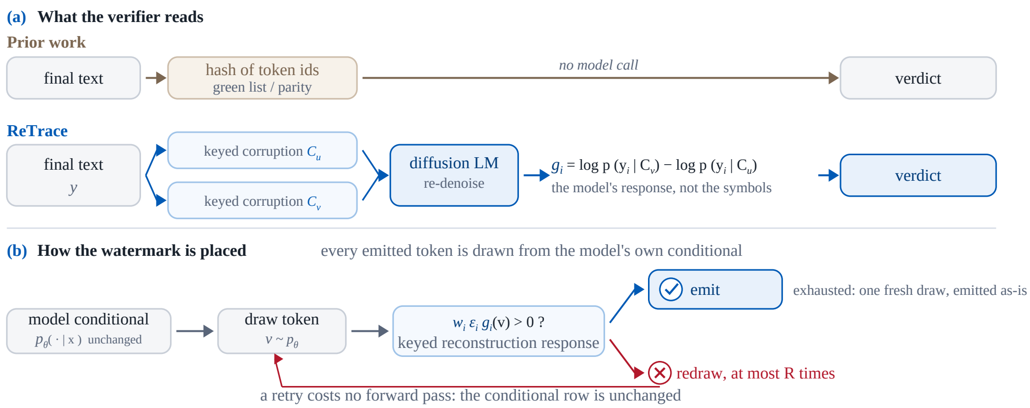

Figure 1: Prior verifiers read token identities; ReTrace's verifier runs the model. **(a)** The finished text is corrupted twice under the key and re-denoised; the contrast between the reconstructions of each observed token is the evidence. **(b)** Embedding never leaves the model's conditional: draw, check the keyed response, redraw at most R times at no forward-pass cost, emit a fresh draw on exhaustion.

improves, a sampler repair that lowered draft perplexity from 22.2 to 9.5 also shrinking the admissible set by 34–60 %. *Better diffusion-model generation reduces post-hoc watermarkability*, and it rules out an entire family of designs.

What survives is to never leave the model's own conditional. Decoding follows the base model's own schedule, and at a keyed *carrier* — a position whose challenge response is genuinely two-sided under the key — the token is drawn from p_θ and simply *redrawn*, at most R times at no forward-pass cost, the conditional row being unchanged, until its reconstruction response agrees with the key; on exhaustion one unconditional draw is emitted as-is. Two guards keep the price bounded. A keyed gate admits only two-sided positions, so positions whose plausible tokens all answer one way fall through to the natural sample and cost nothing; and acceptance can further require a key-free *plausibility floor* $p(v) \geq \kappa\, p_{\max}$, capping any carrier's cost at $-\log \kappa$ beyond the model's own argmax. With the floor in place, raising R buys detection with arithmetic rather than quality: measured perplexity sits *at or below* the matched non-watermarked control (0.74×–0.87×) at TPR@1% of 0.70–0.80 on 512-token documents, and the whole embedder costs 1.28× a plain decode.

ReTrace places the watermark in resampled tokens and reads it back from the model's reconstruction response (Figure 1).

In sum: a watermark read from the model's response rather than its tokens, verifiable from text, model and key alone (§4); a counting statistic with a finite-sample exact binomial null under the ideal-PRF model, false-positive control verified at the *worst* of twenty keys (§4.2); an embedding that never leaves the model's conditional and holds quality at or below its control (§4.3); a mechanism-level negative result — substitution is priced out, and better generation makes it worse (§5); and an evaluation that reports what each method pays, not only what it scores — generation budget and detection cost per row, with the advantages *and* the limitations of the design stated explicitly against them (§6).

2 Related Work

Diffusion language models. Continuous diffusion (Ho et al., 2020) was carried to discrete data by structured corruption processes (Austin et al., 2021) and by score-ratio and masked-diffusion formulations (Lou et al., 2024; Sahoo et al., 2024; Shi et al., 2024; Ou et al., 2025). Instruction-following models at language-model scale followed — LLaDA and its successors (Nie et al., 2025; Zhu et al., 2025), Dream (Ye et al., 2025) — alongside industrial systems built on the same paradigm (Labs et al., 2025). The property we exploit is the training objective itself: these models predict masked tokens from arbitrary surviving context, so they can be applied to a *finished* text and not only used to produce one.

Two further findings from this literature shape our design. First, practical dLLMs are *not* order-invariant even though an ideal conditional predictor would be, and the choice of unmasking order measurably changes what is generated (Kim et al., 2025). Second, decoding is in practice semi-autoregressive: blocks are filled in sequence and diffusion happens within a block (Arriola et al., 2025), and work on accelerating that loop shows that committing a confident token whose left context is still unresolved damages coherence (Wu et al., 2026; Wei et al., 2026; Ben-Hamu et al., 2025). Our embedder inherits both: it steers only *within* the reference block schedule, and it borrows the frontier-anchored commit rule rather than inventing a schedule of its own.

Watermarks for autoregressive models. The dominant family biases token probabilities. Kirchenbauer et al. (2023) seed a green/red vocabulary partition on the preceding token and add a logit bias; detection counts green tokens against a binomial null, and the scheme's reliability and robustness have been studied extensively (Kirchenbauer et al., 2024; Zhao et al., 2023; 2024). A second family avoids distortion by fixing the sampling randomness with a keyed pseudorandom sequence, either through the Gumbel-max trick (Aaronson & Kirchner, 2023; Fu et al., 2024) or with explicit distortion-freeness and undetectability guarantees (Kuditipudi et al., 2024; Christ et al., 2024); others place the signature in model parameters rather than in sampling (Block et al., 2025). Surveys cover the space (Petitcolas et al., 1999; Liang et al., 2026), and the threat model is correspondingly well studied, from paraphrase attacks (Krishna et al., 2023) to extraction of the key itself (Jovanović et al., 2024). What every one of these has in common is the *verification* step: the detector computes a function of the token identities.

Watermarks beyond left-to-right generation. Because the green list is seeded by a prefix that order-agnostic decoding does not provide, transferring the autoregressive constructions to dLLMs requires care. Chen et al. (2025) give a watermark for order-agnostic models by biasing token selection in a way that does not assume a prefix. For diffusion language models specifically, Gloaguen et al. (2026) apply a red–green bias *in expectation* over unresolved context and additionally prefer tokens that make other positions green; Bagchi et al. (2025) use keyed Gumbel-max sampling at each denoising step while preserving the sampling distribution; Wu et al. (2025) and Raban et al. (2026) target the same setting with order-agnostic and efficiency-oriented constructions. Closest to our embedding channel is dgMARK (Hong & No, 2026), which leaves the learned conditionals untouched and instead prefers to unmask positions whose already-preferred token satisfies a keyed parity condition; that discipline is precisely why it is inexpensive, and we adopt it.

What is new here. All of the above differ in *where the watermark is placed* — token probabilities, sampling randomness, decoding order — while agreeing on *what the verifier reads*: a function of the emitted symbols. ReTrace changes the second. The verifier masks part of the finished text under the key, has the model reconstruct it, and compares the reconstructions; the evidence is the model's response, which no token-level statistic can express. The price is equally clear and we report it in every table: those detectors need no model calls, ours needs a fixed number of sequence evaluations. To our knowledge no prior watermark for language models — autoregressive or diffusion — uses the generator itself as the verification oracle in this way.

3 Background and Problem Setup

DLM generation. A diffusion language model reverses a masking corruption: inference fills a masked region over T steps, committing the most confident positions each step, and practical systems decode semi-autoregressively — contiguous blocks left to right, diffusion within the active block (Arriola et al., 2025; Nie et al., 2025). Two consequences matter. The model predicts masked tokens from *arbitrary* surviving context, so it can be applied to a finished text and not only used to produce one; and each emitted token is a sample from a conditional the verifier can reconstruct exactly, since that conditional sees only the finished prefix and masks.

Watermarking problem. An embedder holding a secret key K produces text whose provenance a verifier holding the same key can later establish from the text alone, against three requirements.

Detectability: a test statistic separating watermarked from natural text at a controlled false positive rate. *Quality:* the watermarked distribution should cost little against the same generator without the key. *Verifiability without the trajectory:* detection may use the text, the model and the key, but nothing recorded at generation time. Prior schemes meet these by making the verifier read a function of the token identities — a green-list membership seeded by the preceding token (Kirchenbauer et al., 2023), a parity of the unmasking order (Hong & No, 2026), a keyed sampling sequence (Kuditipudi et al., 2024). This paper asks the verifier to read something else: how the model *reconstructs* the text under keyed corruption. The cost of that choice is a detector that must run the model, and we account for it in every comparison.

4 ReTrace

4.1 The observable

Let y be a generated span and K a secret key. For a block B of y, the key derives L *ablation patterns* $D_1, \dots, D_L$ over the context preceding B, and a pair (u, v) among them. Writing $\mathscr{C}_\ell$ for the context with B, everything after B, and D_ℓ masked, the *challenge contrast* at position $i \in B$ is

$$g_i(w) \;=\; \log p_\theta(w \mid \mathscr{C}_v) - \log p_\theta(w \mid \mathscr{C}_u), \qquad w \in \mathscr{V}. \tag{2}$$

Two properties of $\mathscr{C}_\ell$ matter and neither is optional.

The block and its future are masked in every arm. While B is being decoded, every later block is still `[MASK]`. If the verifier recomputed Eq. (2) on the finished text without re-masking that tail, it would condition on tokens that did not exist when the guidance was produced, and the two sides would be reading different functions. Masking B and everything after it makes the table identical at both times; masking B itself additionally makes it *invariant* to whatever is committed inside B, so a single table serves the block's entire decoding.

The ablations are drawn from the already-final region. $D_\ell \subset \text{prompt} \cup B_{<b}$, which the verifier has verbatim. We use a *contrast* pair — one pattern ablating the context nearest the block, the other the furthest — because recency dominates next-token prediction, so the two arms disagree far more than two random ablations do, widening the dynamic range of g.

4.2 The statistic

Each position carries an independent keyed orientation bit $\varepsilon_i \in \{\pm 1\}$ and a target $w_i \in \{\pm 1\}$, and contributes a bounded indicator

$$m_i \;=\; \mathbf{1}[w_i\, \varepsilon_i\, g_i(y_i) > 0], \qquad T \;=\; \sum_{i \in P} m_i. \tag{3}$$

Proposition 1 (Exact null)**.** *Let the challenge patterns, the per-position pair selection, the carrier set, the group assignment, the tie coins and the orientation bits each be derived from the key under* separate PRF domains*. Fix a text y independent of the key, and condition on the carrier set, the selected pairs, and the challenge magnitudes — all of which are functions of y and the non-orientation domains only. Under the ideal-PRF model the orientation bits are then independent Rademacher variables independent of everything conditioned on, so the m_i are i.i.d.* $\text{Bernoulli}(\frac{1}{2})$ *and $T \sim$* $\text{Binomial}(|P|, \frac{1}{2})$ *exactly.*

The domain separation is load-bearing: the challenge patterns and pair selection are themselves keyed, and sharing PRF output with the orientation bits would let the conditioning event carry information about them. Pair selection uses only the swap-invariant S_i of Eq. (7), so it is orientation-symmetric by construction.

The proposition needs no calibration corpus and no model-specific null estimate, and it is finite-sample rather than asymptotic. Three implementation details are load-bearing.

Ties must be resolved, not dropped. In single precision the emitted token's log-probability rounds to 0 under both arms at a quarter of positions, and scoring the vanished difference as a miss dragged the unwatermarked match rate to 0.32–0.37 against 0.50. Double precision inside the softmax removes the ties; a second keyed coin breaks any residue.

Algorithm 1 LLaDA reference sampler

Input: prompt
Output: text x
1: **for** each block b in order **do**
2: —
3: **while** b has masks **do**
4: pick most confident masked i
5: draw $v \sim p_\theta(\cdot \mid x)$
6: —
7: commit v at i
(no verifier)

Algorithm 2 ReTrace in LLaDA (blue = ours)

Input: prompt, key K, budget R, floor κ
Output: text x
1: **for** each block b in order **do**
2: challenge table for b: masked fwds $\rightarrow g_i$, gate S_i, signs $w_i\varepsilon_i$ (§4.1)
3: **while** b has masks **do**
4: pick most confident masked i
5: draw $v \sim p_\theta(\cdot \mid x)$
6: **if** $S_i > s_{\min}$: redraw ($\leq R$) until $w_i\varepsilon_i g_i(v) > 0 \wedge p_i(v) \geq \kappa p_i^{\max}$; on exhaustion one fresh draw (Eq. 4)
7: commit v at i
Verify: from x: rebuild line-2 tables under K, count $\mathbf{1}[w_i\varepsilon_i g_i(x_i) > 0] \rightarrow \mathrm{Binom}(n, \frac{1}{2})$ (Eq. 3)

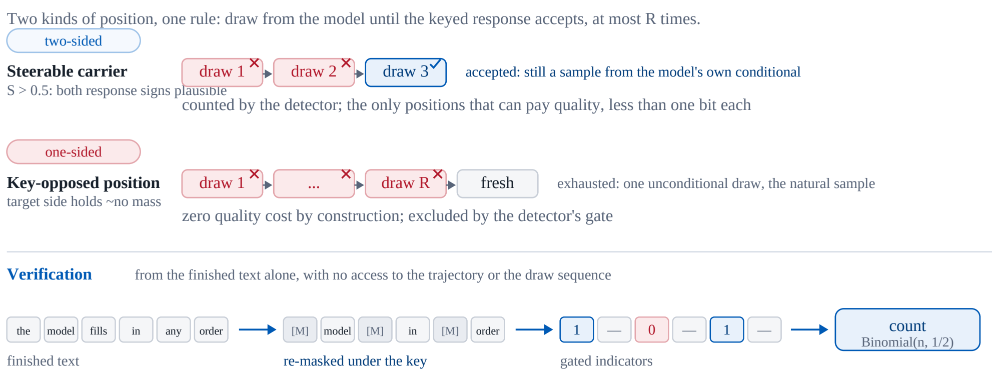

Figure 2: One rule, two regimes. A two-sided carrier accepts within a few redraws — still a sample from the model's conditional, and the only kind of position that pays quality. A key-opposed position rejects every draw and emits the natural sample at zero cost; the gate excludes it. The verifier re-masks under the key and counts gated indicators against an exact binomial null, with no access to the trajectory.

Counting, not averaging. The per-position contrast is heavy-tailed, so an average is dominated by its largest values and sign control has no leverage on it; a count gives every steered position exactly one unit.

Presence, verification and payload are separate tests. Summing hits across payload groups cancels — a balanced message drives half toward m_i=1 and half toward m_i=0, so a perfect watermark sums to $|P|/2$. A presence pool with $w_i \equiv +1$ tests provenance with no claimed message (and never tests a message decoded from the same data); claimed-message verification aligns each group to its bit; payload thresholds each group separately.

4.3 Embedding by Response-Guided Resampling

Generation follows the reference schedule of the base model — blocks in order, diffusion within a block — and commits positions in plain confidence order (Figure 2). Algorithms 1 and 2 put the embedder beside the reference sampler it wraps, line for line: the two differ only in the blue lines — no logit is edited, no commitment is reordered, and deleting the blue lines recovers the base model's decoder exactly. At a carrier position the committed token is chosen by rejection sampling from the model's own conditional:

$$v^{(1)}, v^{(2)}, \ldots \sim p_\theta(\cdot \mid x), \qquad \text{emit the first } v^{(r)} \text{ with } w_i\,\varepsilon_i\,g_i(v^{(r)}) > 0, \quad r \leq R, \tag{4}$$

and if all R guided draws are rejected, one further unconditional draw is emitted as-is. Acceptance may additionally require a *plausibility floor* $p_i(v) \geq \kappa\, p_i^{\max}$, which caps any carrier's cost at $-\log\kappa$ beyond the model's own argmax; the floor is embedder-only and key-free — the detector reads nothing but the emitted token — so the null and the carrier rule are untouched. A retry costs no forward pass — the conditional row is unchanged — so R trades detection strength against arithmetic alone, and the floor keeps that trade from spending the quality budget. Writing A_i for the acceptance set and q_i for its mass, the per-position hit probability and emitted distribution are

$$q_i = \Pr_{v\sim p_i}\left[w_i\,\varepsilon_i\,g_i(v) > 0\right], \qquad \Pr[m_i = 1] \;=\; 1-(1-q_i)^{R+1}, \tag{5}$$

$$v_i \;\sim\; \left(1-(1-q_i)^R\right) p_i(\cdot \mid A_i) \;+\; (1-q_i)^R\, p_i. \tag{6}$$

This *does* condition the sampling distribution on the acceptance predicate and is not claimed to be distortion-free; what makes it cheap is that the support never leaves the model's own conditional, and that the positions where the predicate binds hardest are exactly the positions where it cannot be satisfied at all — which fall through to the natural sample.

Only two-sided positions are carriers. The live acceptance mass is bimodal (quartiles $0.02/0.60/0.98$): at most positions the model's plausible tokens all answer the challenge the same way. Positions are gated by the orientation-symmetric two-sided mass

$$S_i \;=\; 2\min\!\left(q_i^+, q_i^-\right), \qquad q_i^{\pm} = \Pr_{v\sim p_i^{\text{base}}}\left[\pm\, g_i(v) > 0\right], \tag{7}$$

computed under the block-masked conditional: a position with $S_i \leq 0.5$ is not counted, since if the key opposes it retries cannot help and the fallback makes it a certain miss rather than a coin flip. S_i is invariant under swapping the pair's arms and never sees the key's direction or the message, and it is computed from the block-masked conditional, so the verifier reconstructs the identical carrier set from the finished text. The challenge pair itself is chosen per position among the $\binom{L}{2}$ keyed patterns by the same S_i, which raises the mean two-sided mass from 0.28 (a fixed near/far pair) to 0.45.

5 Why Substitution Cannot Pay: the Measurements Behind the Design

The first version of ReTrace embedded by substitution: at keyed carrier positions, replace the model's token with an admissible alternative whose contrast points the right way, subject to a per-token fluency budget τ (a substitution is admissible iff $p(w)/p(w^\star) \geq e^{-\tau}$). It does not work, and the reason generalises beyond our implementation.

There is nothing to substitute. Across 36 configurations spanning three carrier-selection rules, three budgets and two guidance weights, the median carrier position admits *exactly one* token at $\tau = 3$, i.e. $p(w)/p(w^\star) \geq 0.05$, with no truncation of the candidate slate. The best configuration inside a $1.35\times$ perplexity budget reaches TPR@1% $= 0.10$; matched local dgMARK reaches 0.86 at $1.23\times$.

Better generation makes it worse. Our harness initially ranked positions for low-confidence remasking by the maximum of the Gumbel-perturbed logits rather than by $p(x_0)$ under the clean softmax. Repairing it to match the reference sampler moved draft perplexity from 22.2 to 9.5 — and halved the denoiser's entropy, from 0.655 to 0.319, shrinking the admissible set by 34–60 % ($|A(3)|$: $2.5 \to 1.6$; $|A(6)|$: $14.1 \to 5.6$). Part of the early design's apparent strength came from a degraded generator. *Improving a diffusion language model's generation quality reduces its post-hoc watermarkability*, because both quantities are governed by the sharpness of the same conditionals.

Moving to generation time does not rescue substitution. A guidance table valid during generation needs every non-carrier token final first, which forces a two-phase schedule that costs $1.98\times$ perplexity *before any watermark* — more than dgMARK's entire cost. Substitution buys evidence with fluency at a rate set by the model's conditionals, and that rate is unfavourable and worsening; a viable embedding must never override the model's choice, which is what §4.3 does.

Method	Setting	PPL ↓	Ratio ↓	TPR at FPR ↑ 5%	1%	0.1%	Gen. fwd.	Det. fwd.
Budget: one decode per document (Gen. fwd. = tokens generated)								
dgMARK (Hong & No, 2026)	$k=1$, 256 tok	15.40	1.55×	0.88	0.74	0.62	256	**0**
KGW (Kirchenbauer et al., 2023)	$\delta=1$, 256 tok	11.96	**1.03**×	0.97	**0.93**	0.73	256	**0**
KGW (Kirchenbauer et al., 2023)	$\delta=3$, 256 tok	14.05	1.21×	1.00	1.00	1.00	256	**0**
ReTrace V1	best under 1.35× ppl	30.5	1.17×	0.10	0.10	0.00	256	9
Budget: 1.28× *decode — ReTrace V3 (decode + one challenge-table pass per block)*								
ReTrace V3	R=1, 512 tok	11.0	**0.97**×	0.57	0.43	0.20	656	144
ReTrace V3	R=2, 512 tok, held-out	–	1.14×	0.83	0.73	0.63	656	144
ReTrace V3	R=4, 512 tok, held-out	–	1.40×	0.90	0.83	0.77	656	144
ReTrace V3	R=16, κ=0.05, 512 tok	–	**0.74**×	0.90	**0.80**	0.60	656	144
ReTrace V3	R=8, κ=0.1, 512 tok	5.67	**0.87**×	0.80	0.70	0.60	656	144
ReTrace V3	R=8, κ=0.1, 1024 tok	–	**0.83**×	0.81	**0.75**	**0.69**	1312	288
Budget: ∼4× *decode (one-step lookahead per candidate)*								
dgMARK (Hong & No, 2026)	+3-beam, 256 tok	12.23	1.23×	**0.92**	**0.86**	**0.82**	∼1024	**0**

Table 1: Detectability against the quality and compute actually paid, grouped by generation budget. One machine, checkpoint and prompt protocol. *Ratio* is against a non-watermarked control run under the same decoding regime (control PPL: block decoding 9.94, left-to-right 11.61, T=0.8 full-vocab 26.21; ReTrace: paired matched-scheduler control per sub-group); control arms score TPR ≤ 0.03 at the 5% threshold and 0.00 below, $|z| \leq 0.25$. *Gen. fwd.* counts model forwards spent producing the text; ReTrace's tier costs 1.28× a plain decode for its per-block challenge tables, and dgMARK 3-beam sits alone at ∼4× for its lookahead. *Det. fwd.* counts masked evaluations per detection — zero for token-hash detectors, fixed for ReTrace. KGW is scored on distinct bigrams (37% FPR at δ=0 otherwise); its left-to-right regime itself costs 1.17× against block decoding. ReTrace prompt sets are disjoint per sub-group with per-document keys, n=30 per 512-token arm (n=20 for R=16, n=16 at 1024 tokens; quality pairing on valid documents).

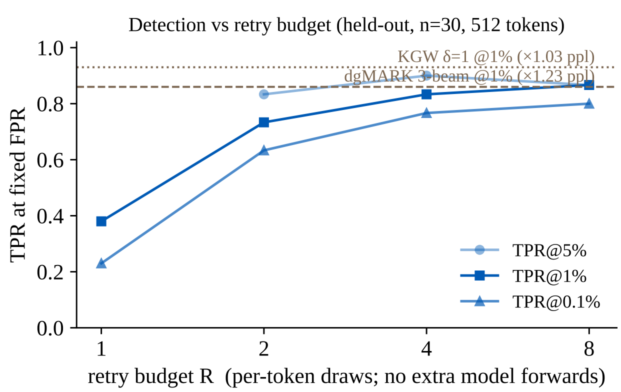

Figure 3: Retries buy detection with arithmetic, but spend the quality budget. n=30 held-out 512-token documents, per-document keys, control at zero false positives; paired quality 1.14 × /1.40 × /1.61× at R=2/4/8. Reference lines are this paper's local baseline reproductions.

6 EXPERIMENTS

One TITAN RTX, `GSAI-ML/LLaDA-8B-Instruct` in fp16; C4 `realnewslike` prompts truncated to 300 characters (dgMARK's protocol); 256–512 tokens at steps = gen_length. Quality is GPT-2-large perplexity against a control *under the same decoding regime* — for ReTrace, the same generator with steering disabled, so the pairing isolates the watermark. Detectability is TPR at fixed FPR, never raw z; development prompts are never reused, and confirmatory runs draw disjoint corpus offsets with per-document keys $K_d = \mathrm{HMAC}(K, \mathrm{nonce}_d)$. Compute is a reported axis, not a footnote: generation forwards appear per row, and the budget-matched comparator for ReTrace (1.28× a plain decode, from one challenge-table pass per block) is dgMARK k=1 — its 3-beam variant spends roughly four decodes per document on lookahead and is flagged as such.

6.1 Main Results and Analysis

The watermark is quality-free, and the earlier evidence against this was a harness artifact. Paired against the matched control, the watermark costs a ΔNLL median of -0.027 at R=1 and -0.037 at R=2 (ratios 0.97×, 0.93×; n=30), replicated on a disjoint held-out set at 0.980× (n=40). Every larger figure previously on record traced to pairing against a differently-scheduled reference generator, and the fingerprint of that confound — R=1 and R=2 showing the same ΔNLL — is what exposed it. The mechanism says this is not luck: at a one-sided position the emitted token is the natural sample whichever way the coin lands, so only genuinely two-sided carriers pay, each less than one bit.

Retries buy detection with arithmetic, not compute — but they do spend the quality budget. On held-out prompts (n=30, 512 tokens), TPR@1% climbs 0.73 → 0.83 → 0.87 across R=2/4/8 while a retry costs no forward pass (Figure 3); the paired quality on the same run is 1.14×, 1.40× and 1.61×. The compute is free; the $-\log q$ budget at two-sided carriers is not, and more retries extract more of it. The operating points that survive comparison are therefore the low-R ones: at R=4 the detection is level with locally reproduced dgMARK 3-beam (0.86) but at higher quality cost (1.40× against 1.23×), so dgMARK dominates that point, whereas nothing in the table reaches the $R \leq 2$ region's quality at any detection strength. Length is the other free axis: at R=1, doubling the document from 256 to 512 tokens lifts 1%-TPR from 0.25 to 0.40 purely by growing the presence count, the same length–power trade dgMARK reports.

A plausibility floor decouples R from the quality budget. Requiring $p(v) \geq \kappa\, p_{\max}$ at acceptance caps each carrier's cost at $-\log \kappa$, and at κ=0.1 it converts the R=8 point from 1.61× to **0.87**× while keeping TPR@1% = 0.70 (rate 0.620 against an unfloored 0.711; control null clean at 0.501). Repetition sits at parity with the control (0.33 against 0.36 on decoded text under the shared validity rule; an earlier raw-token measurement that suggested otherwise was counting EOS padding as repetition). Sub-1.0 ratios deserve suspicion rather than celebration — the floor tilts toward higher-probability tokens, which perplexity rewards — which is why repetition is reported next to the ratio rather than instead of it. Doubling the document to 1024 tokens moves the same configuration to 0.75 at 1% and 0.69 at 0.1% (rate 0.635, ratio 0.83×), with the caveats stated rather than averaged away: repetition opens a +0.06 gap over the control at this length, and only half the documents clear the validity rule for quality pairing. Calibrated and weighted detector variants were also evaluated on identical caches and buy at most 0.02 at 1% — the count is essentially optimal at these rates, so remaining power must come from the acceptance rate and the carrier count, not the statistic.

The simplest detector is the right one. On identical held-out caches the gated count, the gated likelihood ratio and an all-position likelihood ratio are indistinguishable (0.38 at 1% each at R=1), ungated counting is strictly worse (0.33), and every variant holds its null on the control arm. Likelihood weighting buys nothing here because its extra evidence — one-sided positions failing *visibly* — only materialises when the embedder has tried and been refused, which takes retries; at the operating points that matter the deployed detector is the gated indicator with its exact binomial null and no Monte-Carlo machinery.

False positive control holds at the worst key. Twenty keys × 120 non-watermarked generations give 2400 null draws under the deployed decision rule: empirical FPR is at or below nominal at every level *per key*, the worst key included (0.058 at α=0.10, 0.025 at 0.05, 0.000 at 0.01; Table 2), and the matched control arms of every later run sat at zero false positives. An earlier revision reported a Gaussian tail here and measured 0.145 at nominal 0.10; the exact construction is what allows the guarantee to be stated per document rather than on average.

Robustness is the weak axis, and the mechanism says why. Same-model re-denoising of 10% of positions collapses R=1 detection from 0.35 to 0.05 at 1% (deletion likewise), while the attacked control stays at α — the attack erases evidence rather than triggering the detector. The fragility is structural: every block's challenges condition on all earlier blocks, so one edit perturbs the contrast of every later position. Whether the higher rates at R=4–8 buy usable headroom is measured on the saved held-out outputs, and the honest current statement is that token-hash watermarks keep an advantage under heavy post-editing.

Nominal α	Mean FPR	Worst-key FPR	Verdict
0.10	0.0121	0.0583	valid ($\leq \alpha$)
0.05	0.0037	0.0250	valid ($\leq \alpha$)
0.01	0.0000	0.0000	valid ($\leq \alpha$)

Table 2: False positive control holds per key, worst key included. 20 keys $\times$ 120 non-watermarked generations under the deployed rule. Deployment uses one key across documents, so the per-key guarantee is the one that matters; an earlier Gaussian-tail version measured 0.145 at nominal 0.10 and was withdrawn.

Advantages, stated as claims the tables support. (i) *Quality below the control*: with the plausibility floor, ReTrace is the only method in Table 1 operating under 1.0$\times$ (0.74$\times$–0.87$\times$) at nontrivial detection — every baseline pays quality for strength. (ii) *An exact, per-document null*: the decision threshold is $\mathrm{Binomial}(n, \frac{1}{2})$, not an asymptotic z or an empirically calibrated cutoff, and it held at the worst of twenty keys (Table 2). (iii) *The sample never leaves the model*: no logit is edited, no vocabulary is partitioned, no commitment is reordered (Algorithm 2), so failure modes are capped — a key-opposed position falls through to the natural sample at zero cost. (iv) *Near-plain generation budget*: 1.28$\times$ a reference decode, against $\sim$4$\times$ for the strongest dgMARK setting.

Limitations, equally explicit. (i) *Detection needs the model*: $L+1 = 9$ masked sequences per 32-token block — 144 per 512-token document, batchable, seconds on a modern accelerator, but never the hash detectors' zero; this prices a verification API, not a mass-scanning filter, and the method is only proposed for the former. (ii) *Post-editing robustness is the weak axis*: the collapse measured above is structural, and token-hash schemes keep a real advantage under heavy edits. (iii) *Carrier supply depends on model entropy*: sharp conditionals yield few two-sided positions, and the same mechanism that priced out substitution (§5) caps how many carriers a well-trained model offers. (iv) *Detection strength trails the strongest baselines at equal length*: 0.80 at 1% against 3-beam dgMARK's 0.86 and KGW's 0.93 at 256 tokens — the case for ReTrace is the joint operating point, not any single axis.

7 Conclusion

A diffusion language model can be applied to a finished text and not only used to produce one, and ReTrace makes that the verification channel: corrupt the text under a secret key, re-denoise, and read whether the model's reconstruction response agrees with a keyed orientation. Embedding never leaves the model's own conditional — a keyed rejection-resampling with a capped retry budget — and against a matched control it is quality-neutral at the low end (0.93$\times$–1.14$\times$ at $R\leq 2$) and, with a key-free plausibility floor on acceptance, holds sub-control perplexity at high retry budgets: TPR@1% = 0.70 at 0.87$\times$ with repetition at parity (R=8, κ=0.1, held-out). The frontier position is honest rather than dominant — locally reproduced dgMARK (0.86 at 1.23$\times$) still leads on detection, unfloored $R\geq 4$ points are dominated, and nothing in the comparison reaches this quality region at any detection strength. The null is exact per document under the ideal-PRF model, and false positive control holds empirically at the worst of twenty keys.

What separates ReTrace from the watermarks it is compared against is the object verified rather than the strength read: a token-hash detector reads the symbols, ReTrace reads the model's response, and the explicit ledger of what that buys and costs is in §6. The open item on that ledger is post-editing robustness, and hardening the response channel — shorter challenge contexts, redundancy across blocks — is the next thing this line of work must address.

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